

Part A: Ukraine’s Monetary Regime: A Preliminary Analysis

Quantitative Methods in Finance: Final Exam 2025–2026

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1. Documented Chronology of the NBU Exchange Rate Regime (2000–2025)

The chronology below draws on IMF *Annual Report on Exchange Arrangements and Exchange Restrictions* (AREAER) classifications [1], IMF Country Reports on Ukraine [2, 3], official NBU publications [4, 5], and the academic literature. De facto classifications follow the post-2009 IMF methodology (which distinguishes *hard pegs*, *soft pegs*, *managed floating*, and *freely floating* arrangements). Where the de jure and de facto regimes diverge—a pattern extensively documented by Calvo and Reinhart (2002) [11]—the de facto classification is reported.

Capital controls: summary. Significant capital-account restrictions were in place during three main episodes: (i) September 2008–2009, with withdrawal limits and mandatory export-proceed surrender [2]; (ii) February 2014–2019, with progressively tightening then gradually relaxing FX restrictions [8, 9, 7]; and (iii) February 2022–2024, with a full cross-border capital freeze under martial law [10]. These episodes are also noted in the “Key events” column below.

Period	UAH/USD	De facto regime (IMF AREAER)	Key events / capital controls
2000–Aug. 2008	≈ 5.0 (fixed)	Conventional peg to USD <i>AREAER: conventional peg</i>	The NBU defended a hard peg to the US dollar through continuous FX interventions [1]. Inflation remained moderate (5–15%), driven mainly by energy-price pass-through and wage growth. The impossible trinity was fully binding: no independent monetary policy existed in practice. <i>No significant capital controls.</i>
Sep. 2008–Mar. 2009	5.3 $\rightarrow$ 7.9 (+49% UAH/USD; –33% USD/UAH)	Managed float / forced devaluation <i>AREAER: other managed arrangement</i>	Global financial crisis triggered a severe terms-of-trade shock (steel $\approx 40\%$ of exports); capital outflows depleted reserves. The UAH/USD rate rose from 5.3 to 7.9: equivalent to a $\approx 49\%$ nominal depreciation of the hryvnia (or $\approx 33\%$ appreciation of the dollar in UAH terms). Ukraine entered a \$16.4 bn IMF Stand-By Arrangement (November 2008) [2]. <i>Capital controls imposed:</i> withdrawal limits, mandatory surrender of export proceeds.

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Period	UAH/USD	De facto regime (IMF AREAER)	Key events / capital controls
Apr. 2009–Jan. 2014	$\approx 7.9\text{--}8.0$ (de facto fixed)	Stabilised arrangement <i>AREAER: stabilised arrangement</i>	After the 2009 devaluation the NBU re-pegged at ≈ 8 UAH/USD [6]. A 2010 IMF SBA recommended greater flexibility, but the NBU returned to a dollar anchor. <i>Capital controls partially relaxed</i> ; monetary policy autonomy again subordinated to the peg [1].
Feb.–Dec. 2014	$8 \rightarrow 15$ (+88% UAH/USD)	Managed float / sharp devaluation <i>AREAER: floating</i>	Euromaidan crisis, annexation of Crimea (March 2014), and onset of the Donbas conflict triggered massive capital flight. FX restrictions sharply tightened (NBU Resolutions No. 758 and No. 160) [8]. A \$17.1 bn IMF SBA approved April 2014 [3]. <i>Capital controls tightened</i> : mandatory FX surrender, transaction limits.

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Period	UAH/USD	De facto regime (IMF AREAER)	Key events / capital controls
Jan.–Aug. 2015	15 → 30 (+100% UAH/USD)	Managed float under strict capital controls <i>AREAER: floating</i>	Continued sharp depreciation; trough ≈ 30 UAH/USD (February 2015) [7]. Inflation peaked at $\approx 60\%$ YoY (end-2015). NBU raised policy rate to 30% p.a. <i>Capital controls at peak:</i> cash-withdrawal limits, cross-border payment moratorium [9]. <i>Note on cumulative 2014–2015 magnitude:</i> from Feb. 2014 to Feb. 2015 the UAH/USD rate rose from ≈ 8 to ≈ 30 , a $\approx 275\%$ increase: i.e. the hryvnia lost roughly $\approx 73\%$ of its dollar value over the full episode.
Aug. 2015–Jan. 2022	22–28 (gradual de- preciation)	Managed float with inflation targeting <i>AREAER: floating / other managed arrangement (2015–2017), then floating (2018 onward)</i>	NBU Board Resolution No. 541 (18 August 2015) adopted IT as the primary monetary framework [5]. Interim CPI targets: 12 ± 3 pp (end-2016), 8 ± 2 pp (end-2017), 6 ± 2 pp (end-2018), 5 ± 1 pp from 2019 onward [4]. Capital controls progressively relaxed 2016–2019. Ukraine acquired genuine monetary autonomy for the first time in its modern history [6]. <i>Capital controls gradually lifted</i> under IMF conditionality.

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Period	UAH/USD	De facto regime (IMF AREAER)	Key events / capital controls
Feb. 2022–Sep. 2023	29.25 (fixed) → 36.6 (Jul. 2022)	Wartime fixed peg / stabilised arrangement <i>AREAER: stabilised arrangement</i>	Full-scale Russian invasion (24 February 2022): NBU declared martial law and fixed UAH/USD at 29.25 [10]. FX market suspended; cross-border capital flows frozen. NBU financed up to 50% of public deficit (H1 2022). Policy rate raised to 25% (June 2022). Official rate devalued to 36.57 (July 2022) [10]. IMF \$15.6 bn programme approved March 2023. <i>Capital controls at maximum</i> : full cross-border freeze.
Oct. 2023–2025	37 → 44 (managed deprecia- tion)	Managed float (“flexible IT”) <i>AREAER: other managed arrangement</i>	NBU transitioned to managed floating (October 2023), adopting a rate-floor operational framework. Flexible inflation targeting formally adopted in 2024 [6]. CPI fell to 5.1% (end-2023) then recovered to $\approx 7.9\%$ (early 2026) [12]. <i>Capital controls progressively eased</i> under IMF conditionality.

2. Monetary Sovereignty: Genuine vs. Constrained

Ukraine exercised **genuine monetary sovereignty** during only three distinct windows over the 2000–2025 sample: (i) the forced devaluation crisis of 2008–2009, when external shocks compelled the NBU to abandon its hard peg and deploy emergency measures alongside the IMF programme; (ii) the inflation-targeting period of August 2015 to January 2022, during which the hryvnia floated more freely, the NBU actively used its

policy rate to steer inflation toward published targets, and financial-analyst expectations progressively converged toward the 5% goal [6, 4]; and (iii) the partial-recovery phase from October 2023 onward, as capital controls were relaxed and flexible IT was re-established [10]. For the bulk of the sample: the hard-peg plateaux of 2000–2008 and 2009–early 2014, and the wartime fixed rate of February 2022–September 2023: Ukraine’s monetary policy was *de facto* constrained by an exchange-rate anchor, consistent with the “fear of floating” documented by Calvo and Reinhart (2002) [11]: during these phases the UAH/USD rate exhibited near-zero daily volatility while the NBU intervened continuously to defend the peg, leaving interest-rate and money-supply instruments fully subservient to the external anchor.

This asymmetry is **central to the identification strategy in Part B**: Euro Area membership would have represented a *large* treatment during the devaluation episodes (2008–2009, 2014–2015) and during the inflation-targeting period (2016–2022), when Ukraine exercised genuine monetary autonomy and the NBU’s limited credibility generated inflation outcomes that diverged substantially from the Euro Area norm. The credibility-import channel (Barro and Gordon, 1983; Giavazzi and Pagano, 1988) would have been especially potent before 2016, when the NBU lacked a formal inflation target and repeated peg collapses had de-anchored expectations: the gap between actual Ukrainian inflation and the counterfactual Euro Area path should therefore be *largest* in 2008–2009 and 2014–2015, and should narrow after 2016 as the NBU partially converged toward a modern IT framework. Conversely, the counterfactual treatment is *small* during peg plateaux (2000–2008, 2009–2014, 2022–2023), when monetary policy was already subordinated to an external anchor (the dollar rather than the euro). A well-identified counterfactual must therefore exhibit **time-varying treatment intensity**: larger deviations from actual Ukrainian inflation during crisis-devaluation and IT episodes, and smaller deviations during stable-peg periods.

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